

Part c Verification of FastAPI - Pass the data to the API (Request body)

Ensure command below **uvicorn main:api --reload** has run:

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$ lsof -i :8000
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$ uvicorn main:api --reload
INFO: Will watch for changes in these directories: ['/home/ubuntu/datascientest-data-engineering/fast-api']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [1980] using StatReload
INFO: Started server process [1982]
INFO: Waiting for application startup.
INFO: Application startup complete.
```

Modify [main.py](#) file:

```
main.py 2, M X
datascientest-data-engineering > fast-api >
1  from fastapi import FastAPI
2  from typing import Optional
3  from pydantic import BaseModel
4
5
6  api = FastAPI()
7
8  class Item(BaseModel):
9      itemid: int
10     description: str
11     owner: Optional[str] = None
12
13 @api.post("/item")
14 def post_item(item: Item):
15     return {
16         "itemid": item.itemid
17     }
```

Open URL <http://127.0.0.1:8000/docs>

FastAPI 0.1.0 OAS 3.1

openapi.json

default

POST /item Post Item

Schemas

HTTPValidationError > Expand all object

Item > Expand all object

ValidationError > Expand all object

Run following commands in terminal:

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$ curl -X 'POST' -i \
> 'http://127.0.0.1:8000/item'
HTTP/1.1 422 Unprocessable Entity
date: Thu, 12 Jun 2025 15:12:42 GMT
server: uvicorn
content-length: 82
content-type: application/json

{"detail":[{"type":"missing","loc":["body"],"msg":"Field required","input":null}]}ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$
● curl -X 'POST' -i \
> 'http://127.0.0.1:8000/item' \
> -H 'Content-Type: application/json' \
> -d '{
>   "itemid": 1234,
>   "description": "my object",
>   "owner": "Daniel"
> }'
HTTP/1.1 200 OK
date: Thu, 12 Jun 2025 15:12:57 GMT
server: uvicorn
content-length: 15
content-type: application/json

{"itemid":1234}ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$ curl -X 'POST' -i \
> 'http://127.0.0.1:8000/item' \
> -H 'Content-Type: application/json' \
> -d '{
>   "itemid": 1234,
>   "description": "my object"
> }'
HTTP/1.1 200 OK
date: Thu, 12 Jun 2025 15:13:11 GMT
server: uvicorn
content-length: 15
content-type: application/json

{"itemid":1234}ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$ curl -X 'POST' -i \
> 'http://127.0.0.1:8000/item' \
> -H 'Content-Type: application/json' \
> -d '{
>   "itemid": 12345
> }'
HTTP/1.1 422 Unprocessable Entity
date: Thu, 12 Jun 2025 15:13:24 GMT
server: uvicorn
content-length: 108
content-type: application/json

{"detail":[{"type":"missing","loc":["body","description"],"msg":"Field required","input":{"itemid":12345}}]}ubuntu@ip-172-31-18-113:~/datascientest-
data-engineering/fast-api$ curl -X 'POST' -i \
> 'http://127.0.0.1:8000/item' \
> -H 'Content-Type: application/json' \
> -d '{
>   "itemid": 12345,
>   "description": "my object",
>   "other": "something else"
> }'
HTTP/1.1 200 OK
date: Thu, 12 Jun 2025 15:13:35 GMT
server: uvicorn
content-length: 16
content-type: application/json

{"itemid":12345}ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$
```

Modify [main.py](#) file:

```
main.py 2, M X
datascientest-data-engineering > fast-api >
1  from fastapi import FastAPI
2  from typing import Optional
3  from pydantic import BaseModel
4
5
6  api = FastAPI()
7
8  class Item(BaseModel):
9      itemid: int
10     description: str
11     owner: Optional[str] = None
12
13     @api.post("/item")
14     def post_item(item: Item):
15         return {"itemid": item.itemid}
```

```

ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$ curl -X 'POST' -i \
> 'http://127.0.0.1:8000/item' \
> -H 'Content-Type: application/json' \
> -d '{
>   "itemid": 1234,
>   "description": "my object",
>   "other": "not in schema"
> }'
HTTP/1.1 500 Internal Server Error
date: Tue, 17 Jun 2025 12:24:26 GMT
server: uvicorn
content-length: 21
content-type: text/plain; charset=utf-8

Internal Server Errorubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$

```

Modify [main.py](#) file again:

```

main.py 2, M X
datascientest-data-engineering > fast-api >
1  from fastapi import FastAPI
2  from typing import Optional
3  from pydantic import BaseModel
4
5
6  api = FastAPI()
7
8  class Item(BaseModel):
9      itemid: int
10     description: str
11     owner: Optional[str] = None
12
13     @api.post('/item')
14     def post_item(item: Item):
15         return item

```

```

Internal Server Errorubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$ curl -X 'POST' -i \
> 'http://127.0.0.1:8000/item' \
> -H 'Content-Type: application/json' \
> -d '{
>   "itemid": 12345,
>   "description": "my object",
>   "other": "something else"
> }'
HTTP/1.1 200 OK
date: Tue, 17 Jun 2025 12:28:05 GMT
server: uvicorn
content-length: 55
content-type: application/json

{"itemid":12345,"description":"my object","owner":null}ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$

```

```

{"itemid":12345,"description":"my object","owner":null}ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$ curl -X 'P
OST' -i \
> 'http://127.0.0.1:8000/item' \
> -H 'Content-Type: application/json' \
> -d '{
>   "itemid": 12345,
>   "description": "my object",
>   "owner": "Daniel",
>   "other": "something extra"
> }'
HTTP/1.1 200 OK
date: Tue, 17 Jun 2025 12:38:40 GMT
server: uvicorn
content-length: 59
content-type: application/json

{"itemid":12345,"description":"my object","owner":"Daniel"}ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$

```